

ESP32 INDUSTRIAL I/O CONTROLLER

Hardware Datasheet & User Manual

Industrial 24 V field-I/O controller with ESP32-WROOM-32E, USB service, RS485, analog inputs, regulated power rails and relay output.

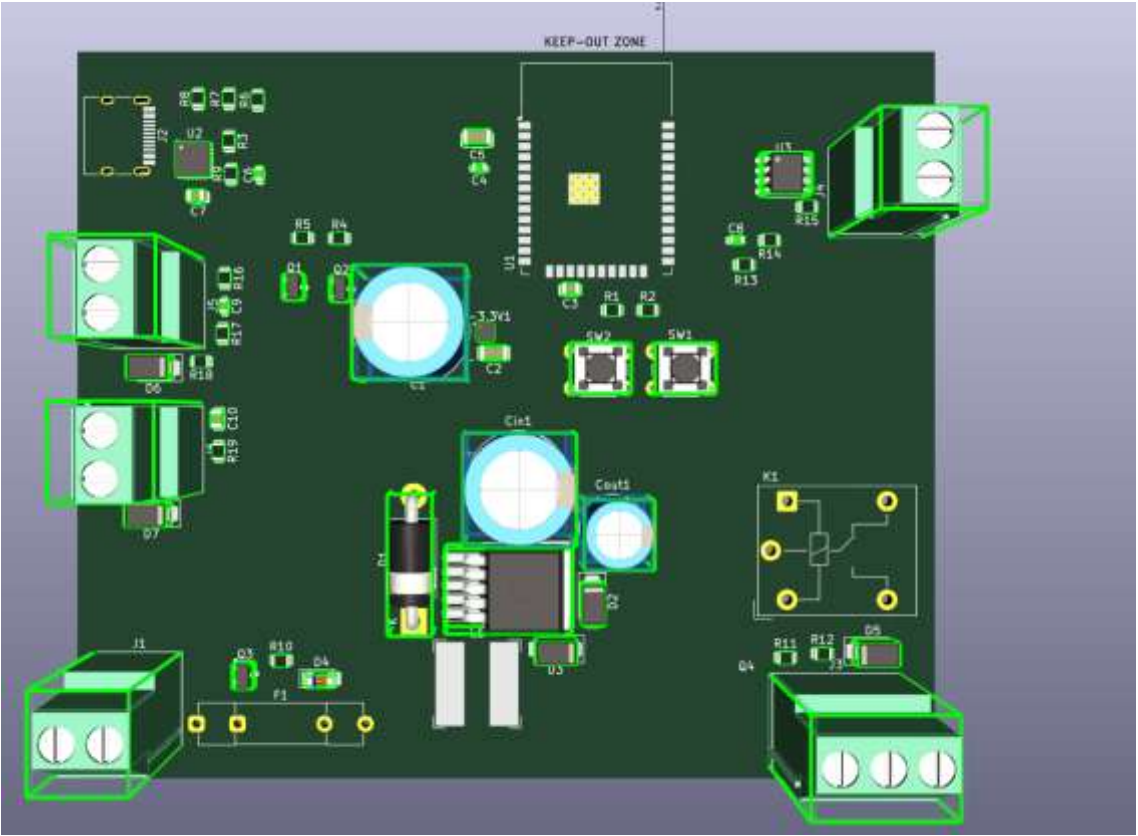


Figure 1. Final PCB 3D view (engineering prototype, Hardware Revision A).

DOCUMENT	VALUE
Document ID	EIC-4L-DAT-001
Hardware Revision	Rev A - Routed PCB
Document Revision	1.0
PCB	4-layer FR-4, nominal 1.6 mm
Primary Controller	ESP32-WROOM-32E
Intended Use	Industrial field-I/O controller / prototyping / engineering evaluation

DOCUMENT STATUS

Project-specific engineering datasheet and user manual; not an Espressif product document or certification.

Contents

Datasheet-style technical reference and operating guide

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Revision History

REV	DATE	DESCRIPTION
1.0	2026-09	Professional datasheet + user manual format prepared from the completed KiCad design record and project documentation.

DOCUMENT CONVENTION

Connector designators, net names and component references are kept consistent with the project design record. Where a production limit was not established during the project, it is explicitly identified as a validation item.

1. Product Overview & Features

What the board is and what the user can connect

The ESP32 Industrial I/O Controller is a 24 V DC field interface board built around the ESP32-WROOM-32E. It combines analog process inputs, RS485 communication, USB service/programming, regulated power conversion and a relay output in a single 4-layer PCB.

Key Features

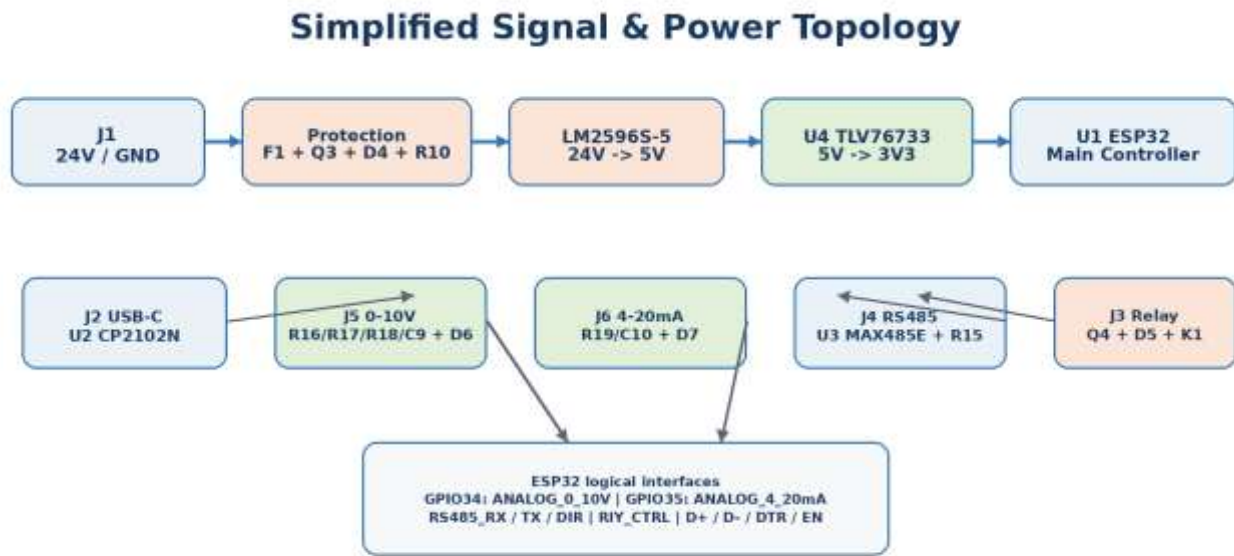
FEATURE	IMPLEMENTATION
Controller	ESP32-WROOM-32E control core with antenna keep-out at the board edge.
Field Power	24 V DC nominal input at J1 with a 500 mA fuse and input protection stage.
Analog I/O	0-10 V input at J5 and 4-20 mA input at J6, with signal conditioning and protection.
Communication	RS485 A/B interface at J4 using MAX485E with 120 ohm termination.
Service	USB-C service/programming interface at J2 using CP2102N USB-UART.
Output	24 V SPDT relay K1 with transistor driver, flyback protection and COM/NC/NO connector J3.
Power Rails	LM2596S-5 buck stage for 5 V and TLV76733 regulator for 3.3 V.
PCB	4-layer stack with continuous inner GND plane and dedicated power distribution layer.
Mechanical	Perimeter field connectors with user-facing silkscreen labels for installation.

5 A LOAD TARGET

The project record identifies an external 24 V DC inductive motor/load up to 5 A as a design target for the relay contacts. Final field-safe current depends on the selected relay, terminal, copper, temperature rise, wiring and load inrush; validate before production use.

2. Functional Architecture

Major power, control and field-interface domains



Functional topology only; the project KiCad schematic remains the authoritative pin-level design source.

Figure 2. Simplified signal and power topology (functional schematic).

Functional Description

DOMAIN	FUNCTION
24 V input	J1 receives the nominal 24 V field supply; the input protection stage feeds the protected field rail.
Power conversion	LM2596S-5 generates the 5 V rail. The TLV76733 stage provides the 3.3 V domain.
Control	ESP32-WROOM-32E executes firmware and interfaces with the analog, RS485, USB and relay domains.
USB service	J2 connects to CP2102N for USB-UART service/programming.
Analog	J5 and J6 provide conditioned 0-10 V and 4-20 mA field inputs.
RS485	MAX485E provides the A/B field bus interface at J4.
Relay	ESP32 control drives Q4 and K1; J3 exposes COM/NC/NO relay contacts to the external load.

3. Board Pinout & User Interfaces

Front-panel reference for installation and service

Board Pinout & User Connection Map

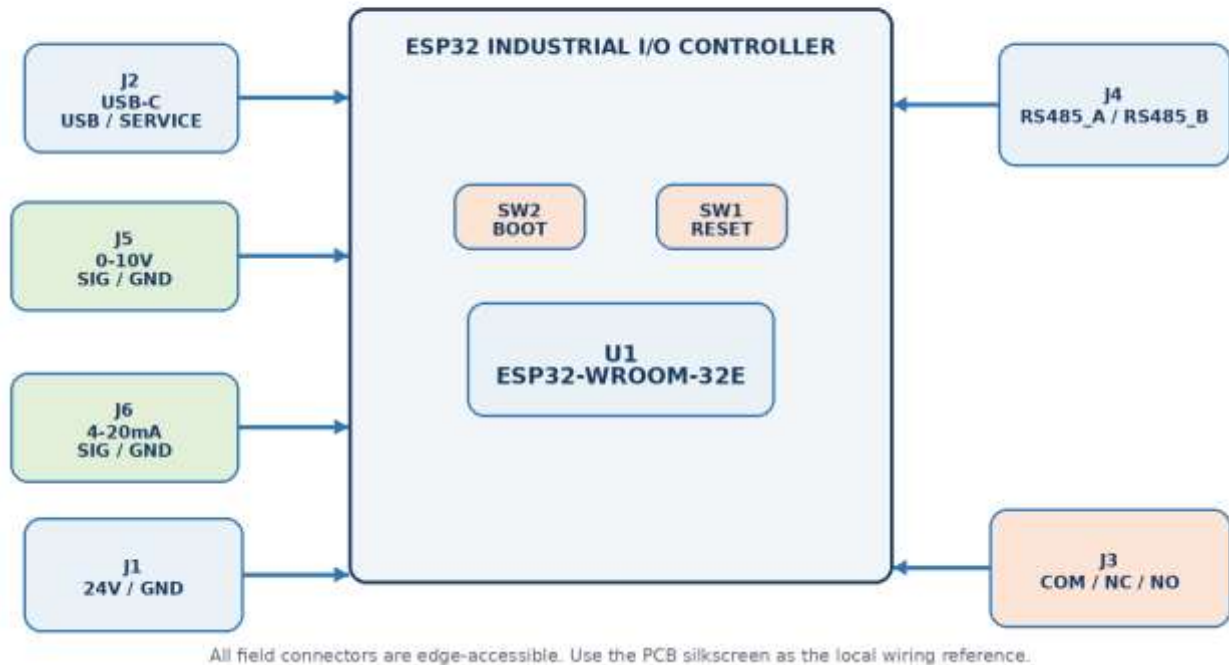


Figure 3. Board pinout and user connection map.

Connector / Control Pinout

DESIGNATOR	TYPE	PIN / TERMINAL	USER FUNCTION
J1	2-pin screw terminal	1: 24V 2: GND	24 V DC board power
J2	USB-C	USB	Service / programming / UART bridge
J5	2-pin screw terminal	SIG / GND	0-10 V analog input
J6	2-pin screw terminal	SIG / GND	4-20 mA analog input
J4	2-pin screw terminal	A / B	RS485 differential bus
J3	3-pin screw terminal	COM / NC / NO	Relay-switched external circuit
SW1	Pushbutton	RESET	ESP32 reset
SW2	Pushbutton	BOOT	Bootloader / flash entry

FIELD WIRING

Use the silkscreen labels on the PCB as the primary wiring reference. Power down the installation before wiring field terminals.

4. Electrical & Interface Specifications

Design-level specifications captured from the project

PARAMETER	VALUE / IMPLEMENTATION	STATUS
Input supply	24 V DC nominal	Design value
Input fuse	F1 = 500 mA	Implemented
Protected rail	24V_PROTECTED	Implemented
5 V converter	LM2596S-5 buck stage	Implemented
3.3 V regulator	TLV76733DRVx	Implemented
0-10 V input	J5; divider/filter/protection; GPIO34	Implemented
4-20 mA input	J6; 150 ohm shunt; filter/protection; GPIO35	Implemented
RS485	MAX485E, A/B, 120 ohm termination	Implemented
Relay	24 V SPDT, JQC-3FF-024-1Z family	Implemented
External load target	24 V DC, up to 5 A inductive	Target / validate
PCB	4 layers, nominal 1.6 mm	Implemented

Key Component Reference

REF	PART	ROLE
U1	ESP32-WROOM-32E	Main controller
U2	CP2102N	USB-UART bridge
U3	MAX485E	RS485 transceiver
U4	TLV76733DRVx	3.3 V regulator
Buck1	LM2596S-5	5 V buck stage
K1	JQC-3FF-024-1Z family	24 V SPDT relay
Q4	BSS138	Relay low-side driver

SPECIFICATION SCOPE

This table is a design-level reference, not a guaranteed production datasheet. Absolute ratings, environmental ratings, EMC/ESD performance and field current capability must be validated against the selected components and the final manufactured PCB.

5. Analog Inputs

0-10 V and 4-20 mA process-signal interfaces

5.1 0-10 V Input (J5)

ITEM	IMPLEMENTATION
Connector	J5: SIG + GND
Protection	D6 clamp/protection device
Divider	R16 = 10 kOhm; R17 = 20 kOhm
Filter / conditioning	R18 = 10 kOhm; C9 = 100 nF
MCU input	GPIO34 / ANALOG_0_10V

The J5 signal is conditioned by the resistor network, filtering and the D6 protection device before reaching the ESP32 ADC input. The project record intentionally does not assign a final calibrated transfer function until the final resistor orientation and schematic are verified.

5.2 4-20 mA Input (J6)

ITEM	IMPLEMENTATION
Connector	J6: SIG + GND
Protection	D7 clamp/protection device
Shunt	R19 = 150 Ohm
Filter	C10 = 100 nF
MCU input	GPIO35 / ANALOG_4_20mA
Nominal shunt levels	4 mA -> 0.6 V; 20 mA -> 3.0 V

INPUT PRACTICE

Keep sensor wiring away from the buck switching node and relay/motor-current path. Use the same field reference convention as the connected transmitter.

6. RS485 Interface

Field bus connection and termination

ITEM	IMPLEMENTATION
Transceiver	U3 = MAX485E
Connector	J4 = RS485_A / RS485_B
Termination	R15 = 120 Ohm across A/B
Bias network	R13 = 10 kOhm; R14 = 20 kOhm in the implemented interface network
Decoupling	C8 = 100 nF
MCU-side logical nets	RS485_RX, RS485_TX, RS485_DIR

User Connection

Connect the external differential pair to J4 using the on-board labels RS485_A and RS485_B. Maintain consistent A/B polarity across the network and use appropriate twisted-pair field cabling.

PCB Routing Intent

RULE	IMPLEMENTATION
Pairing	Keep A and B short and routed together on the signal layer.
Termination	Place R15 close to the A/B interface region.
Grounding	Use the continuous In1.Cu GND plane; do not create arbitrary ground splits.
Noise separation	Keep RS485 routing away from the LM2596 switching loop and relay/motor-current path.

CONNECTOR SCOPE

The current project design exposes A/B on a 2-pin J4 connector. A separate signal-reference conductor is not provided by J4 in the current design.

7. Relay Output

24 V SPDT control and external load interface

ITEM	IMPLEMENTATION
Relay	K1 = JQC-3FF-024-1Z family, 24 V coil, SPDT
Driver	Q4 = BSS138 low-side switch
Driver network	R11 = 100 Ohm; R12 = 10 kOhm
Flyback	D5 across relay coil
Field connector	J3 = COM / NC / NO
External load path	External circuit connects through K1 contacts and J3

User Connection

J3 exposes the relay contacts as COM, NC and NO. Wire the external circuit according to the required relay state. The relay coil driver is a control circuit; the external load current flows through the relay contacts and field wiring.

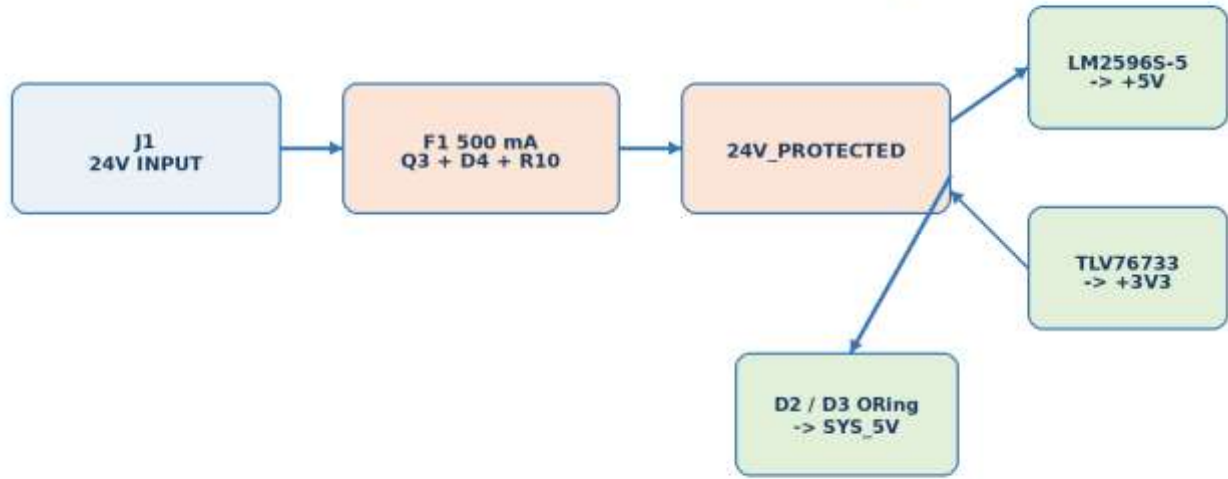
5 A PATH BOUNDARY

The external 5 A motor/load path is not routed through Q4, D5, R11 or R12. Those parts only control the relay coil. Final contact current and temperature-rise validation are required for an inductive load.

8. Power System & Protection

24 V input, 5 V conversion and 3.3 V regulation

Power Tree & Rail Relationships



24 V field input -> protection -> 5 V conversion / source selection -> 3.3 V logic

Figure 4. Power tree from field input to regulated rails.

BLOCK	COMPONENTS	FUNCTION
Input protection	J1, F1 500 mA, Q3 PMOS, D4, R10	Field input fuse / protection / protected rail
Buck input	Cin1 = 680 uF	Bulk energy at LM2596 input
Buck stage	LM2596S-5, D1 = 1N5820, L1 = 33 uH	24V_PROTECTED -> 5 V
Buck output	Cout1 = 220 uF	5 V output bulk filtering
5 V source selection	D2/D3 = SS34	+5V / USB VBUS OR-ing to SYS_5V
3.3 V regulator	U4 TLV76733DRVx, C1/C2 = 10 uF	SYS_5V / 5 V domain -> 3.3 V

LM2596 PLACEMENT / ROUTING

The high-frequency buck loop is kept compact on the component side. Use short connections around the switch/diode/inductor/input-capacitor region and connect GND into the continuous inner plane.

9. PCB Architecture & Routing

4-layer board strategy

4-Layer PCB Stackup



Nominal finished board thickness: 1.6 mm

Figure 5. 4-layer stackup and intended layer functions.

LAYER	PRIMARY FUNCTION
F.Cu / L1	Main signal routing, local power, critical buck loop
In1.Cu / L2	Continuous GND plane
In2.Cu / L3	Power distribution and selected routing
B.Cu / L4	Secondary signals and additional copper for high-current paths

Routing Priority

ORDER	ROUTE FIRST / LAYER INTENT
1	LM2596 switching loop - F.Cu
2	24 V protected power distribution - F.Cu then In2.Cu
3	J3 <-> K1 relay-contact high-current path - F.Cu/B.Cu
4	5 V / 3.3 V distribution - In2.Cu + short F.Cu local links
5	USB D+/D- - F.Cu
6	RS485 A/B - F.Cu
7	Analog inputs - F.Cu
8	Relay control - F.Cu
9	Remaining ESP32 signals - F.Cu/B.Cu

10. Installation & Wiring

Field connection procedure

Before Power-On

CHECK	ACTION
1	Inspect the PCB for damage, foreign objects and loose hardware.
2	Verify J1 polarity: 24V and GND.
3	Confirm all field connectors are tightened and correctly labeled.
4	Keep the ESP32 antenna region clear of metal/enclosure structures that intrude into the designated keep-out.

Field Wiring

INTERFACE	CONNECT
J1 - Power	24 V DC supply -> 24V and GND
J5 - 0-10 V	Process signal -> SIG; reference -> GND
J6 - 4-20 mA	Loop signal -> SIG; reference/return -> GND as required by transmitter topology
J4 - RS485	Differential pair -> A and B
J3 - Relay	External load circuit -> COM / NC / NO
J2 - USB	USB-C service cable

SAFETY

Power down the installation before wiring field terminals. This manual is an engineering reference and does not replace machine-level electrical design, protective-device coordination, applicable wiring standards or load-manufacturer instructions.

11. Programming / Service

USB-C and ESP32 boot/reset workflow

USB Service Interface

J2 provides the USB-C connection. U2 (CP2102N) acts as the USB-UART bridge to the ESP32 service/programming interface defined in the design.

CONTROL	USE
J2 USB-C	Connect the development PC / service computer.
SW2 BOOT	Use for bootloader / flash entry when required by the firmware workflow.
SW1 RESET	Reset the ESP32 controller.
CP2102N	Provides the USB-UART bridge exposed by J2.

Generic Service Sequence

STEP	ACTION
1	Connect USB-C to J2.
2	Apply the intended board power arrangement if required by the firmware/service setup.
3	Use SW2 (BOOT) and SW1 (RESET) according to the firmware flashing procedure.
4	Verify the computer detects the USB-UART interface.
5	Program or service the ESP32 using the project firmware tools.
6	After programming, release BOOT and reset the controller for normal operation.

SOFTWARE DEPENDENCY

COM-port enumeration, drivers, bootloader behavior and firmware commands are dependent on the host operating system and the firmware loaded into the ESP32.

12. Operating Procedure

Normal user sequence

STEP	USER ACTION
1	Inspect the board and wiring.
2	Connect 24 V DC to J1 following the 24V/GND labels.
3	Connect process signals at J5 and/or J6 as required.
4	Connect the RS485 network at J4 if used.
5	Connect the external circuit to J3 using COM/NC/NO if relay control is required.
6	Apply power and allow the firmware to initialize.
7	Use USB-C service access only when required for configuration, programming or diagnostics.
8	For reset, press SW1. For bootloader entry, use SW2 with the firmware-defined reset sequence.

Shutdown

Remove the field supply before disconnecting field wiring. For inductive loads, allow the external machine/system to reach its safe shutdown state before removing power.

13. Troubleshooting & Maintenance

First checks for common symptoms

SYMPTOM	FIRST CHECKS
No power	J1 polarity -> F1 -> Q3/input protection -> 24V_PROTECTED -> buck input
ESP32 not detected over USB	USB-C cable -> CP2102N area -> host COM/driver -> BOOT/RESET sequence
0-10 V reading wrong	J5 wiring -> divider values/orientation -> D6/filter -> GPIO34 scaling
4-20 mA reading wrong	J6 wiring -> 150 Ohm shunt -> loop integrity -> GPIO35 scaling
RS485 no communication	A/B polarity -> baud/configuration -> termination/bias -> field cable
Relay does not switch	24V_PROTECTED -> Q4 control -> D5/coil -> J3 contact wiring
Resets under load	24 V supply droop -> buck output -> relay/load transients -> return path

Maintenance Checklist

PERIODIC CHECK	WHAT TO LOOK FOR
Field terminals	Tightening, discoloration, mechanical damage
Relay / load wiring	Heating, discoloration, contact stress on inductive loads
Power rails	24 V input, 5 V rail and 3.3 V rail within expected design values
Analog inputs	Sensor wiring, shunt/divider condition and calibration
Antenna region	No metal or enclosure intrusion into the keep-out

14. Developer Reference

Logical nets and implementation references

LOGICAL NET / NODE	USE
ANALOG_0_10V	GPIO34 analog measurement path from J5
ANALOG_4_20mA	GPIO35 analog measurement path from J6
RS485_RX	MCU receive path from MAX485E
RS485_TX	MCU transmit path to MAX485E
RS485_DIR	RS485 direction control
RIY_CTRL	Relay-driver control net in the current design record
D+ / D-	USB data pair for CP2102N service interface
DTR / EN	USB-UART service/reset-related path in the schematic
+3V3	Main 3.3 V logic rail
+5V / SYS_5V	5 V distribution / selected source rail
24V_PROTECTED	Protected field-supply rail

Selected Footprints

REF	SELECTED / PROJECT FOOTPRINT
J1/J4/J5/J6	TerminalBlock_Phoenix:TerminalBlock_Phoenix_MKDS-3-2-5.08_1x02_P5.08mm_Horizontal
J3	TerminalBlock_Phoenix:TerminalBlock_Phoenix_MKDS-3-3-5.08_1x03_P5.08mm_Horizontal
K1	Relay_THT:Relay_SPDT_Hongfa_JQC-3FF_0XX-1Z
L1	Inductor footprint selected for 33 uH design component
R1-R19	Resistor_SMD:R_0805_2012Metric

15. Design Notes, Limits & Validation

What is implemented vs what must still be validated

Implemented Design Decisions

AREA	DECISION / STATUS
PCB stack	4 copper layers; nominal 1.6 mm board
Grounding	In1.Cu used as a continuous GND plane
Power	In2.Cu used for power distribution; F.Cu used for local/critical power paths
Connectors	Field connectors selected with mechanical 3D models and placed at board perimeter
Antenna	ESP32 antenna region retained at board edge with keep-out
Routing	Critical power, USB, RS485, analog and relay paths routed and checked

Validation Items Before Production Use

ITEM	WHY IT NEEDS VALIDATION
External 5 A inductive load	Relay contact rating, inrush, copper temperature rise, connector and wiring capability
0-10 V calibration	Final divider orientation and firmware scaling
4-20 mA field topology	Transmitter loop arrangement and reference convention
USB / boot workflow	Host OS, driver and firmware behavior
EMC / ESD	System-level qualification not established by PCB DRC alone
Environmental performance	Temperature, humidity, enclosure and vibration limits not established in this project record

ENGINEERING BOUNDARY

Passing KiCad DRC and Gerber inspection confirms design-rule compliance for the configured PCB checks; it does not by itself establish electrical safety, EMC compliance, thermal capability, or product certification.

Appendix A. Connector Quick Reference

One-page field wiring card

CONNECTOR	LABELS / PINOUT	PURPOSE
J1	24V GND	24 V DC board supply
J2	USB	Service / programming
J5	SIG GND	0-10 V input
J6	SIG GND	4-20 mA input
J4	RS485_A RS485_B	RS485 bus
J3	COM NC NO	Relay contacts
SW2	BOOT	Bootloader entry
SW1	RESET	Controller reset

Power-Up Quick Checklist

J1 polarity correct -> field connectors correct -> USB optional -> apply 24 V -> verify controller behavior -> verify signals/RS485/relay function.

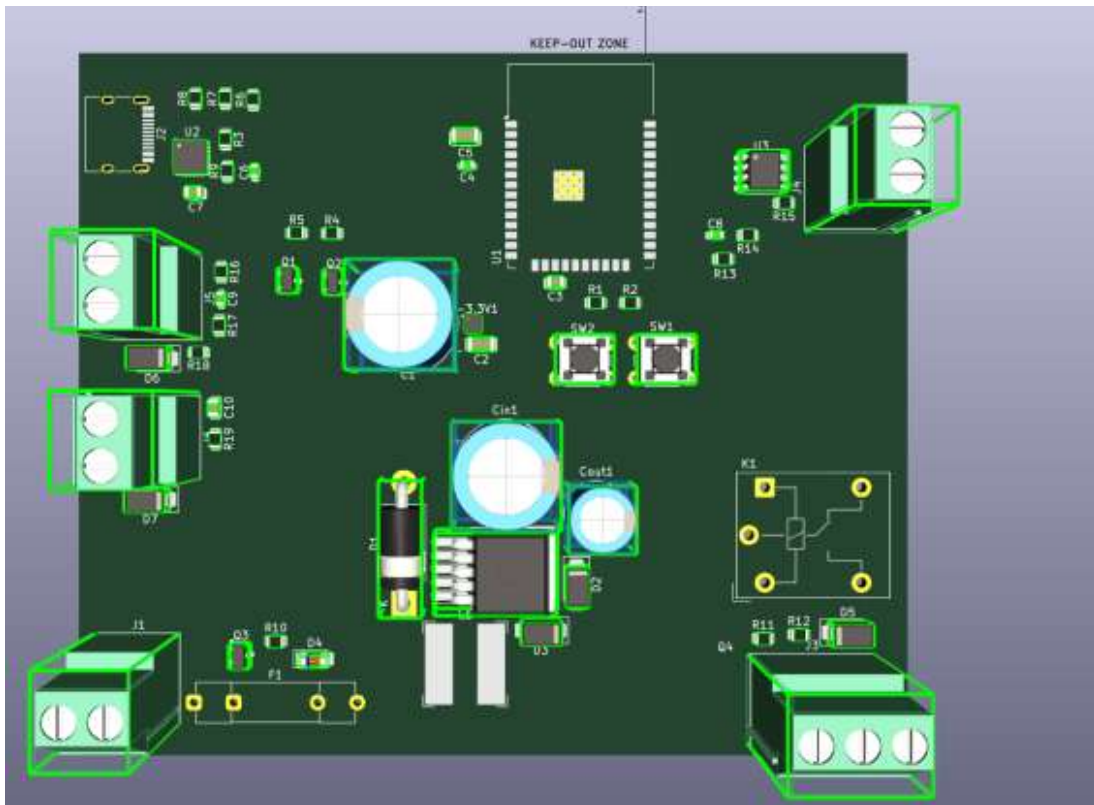


Figure 6. Reference 3D view for connector and board orientation.

Appendix B. Revision History & Project File Set

Documentation package for GitHub release

Recommended GitHub Repository Structure

PATH / FILE	PURPOSE
README.md	Project overview, goals, screenshots and quick start
hardware/	KiCad schematic, PCB and project files
docs/ESP32_Industrial_IO_Controller_Datasheet_User_Manual.pdf	Published user-facing document
docs/ESP32_Industrial_IO_Controller_Design_Process.pdf	Engineering process / design history
gerbers/	Fabrication outputs
assets/	3D renders, architecture and interface images
firmware/	Firmware source and release notes when available

Revision History

REV	DATE	CHANGE
1.0	2026-09	Initial professional datasheet/user-manual release based on Hardware Revision A project record.

END OF DOCUMENT

For project source, fabrication data and the engineering process record, use the companion documentation in the GitHub repository.